Henry Wei
October 8 2026

Formula Slug Simulation Onboarding

At first when looking through the onboarding project I noticed that all the parts used the same step by step method outlined by the template. Because of this, I wanted a clear understanding of the template and polish out the first simulation so the following ones would be easier. I was somewhat confused by the template, specifically the animate library, so I first programmed a simplified test file using while loops and printed the calculations into the terminal. Once, I double checked some of the calculations and knew that the calculation component of the simulation worked, I drew out a simple flow chart of the template to understand how the state dataclass and the step and animate functions would work together. Using this chart I plugged in the logic from my test file and got part one running and graphing the expected results.

However, there were a few improvements I wanted to make before I moved on to the following parts. Specifically, the throttle input function. I was using a linear function to get from 0% throttle to 100% throttle which demonstrated that the simulation works but the program couldn't handle varying inputs. From my experience as a sim racer and looking at telemetry, I knew that inputs are often smooth and avoid erratic inputs that can destabilize the car. At first, I tried using sine functions to match the requirement of reaching 100% throttle over seven seconds while keeping the inputs smooth. While this worked, when listening to Nathaniel's explanation of cubic spline interpolations, I realized that using cubics would be both easier to calculate and allow for more diverse functions.

After getting the first part down, other than a few clarifications, the following parts went pretty smoothly. My main takeaway from this project is the importance of simplifying larger problems into small parts and understanding fundamental ideas before tackling the entire problem.

(No space for improvements so they will be in presentation)